



Telrad Foundation provides remote radiology reads for the Ramakrishna Mission hospital in Itanagar, Arunachal Pradesh

‘medical jewellery.’

These devices are powered by advanced signal processing and have sophisticated chip radios—they combine radio frequency (RF) capabilities along with processing and other required features. These can be connected through low-power Bluetooth or other technologies to a handset. Zigbee, a Wi-Fi like wireless solution, is being touted to be the ideal low-power local-area networking (LAN) solution for health monitoring solutions. The Continua Healthcare Alliance, an industry coalition working towards standards for telemedicine and health monitoring, also recommends Zigbee. It defines certain standards for the wide-area network (WAN) used for health monitoring, etc.

The handset could then transmit the sensor data to a central location, using a WAN, like a mobile network. This information would then be captured and interpreted by a server, to aid doctors in their treatment. This way, the doctors can get hold of a ‘trend’ rather than a mere ‘reading’ of the patient’s medical condition.

However, there is a lot of work to be done by the industry before such mobile-based health monitoring solutions become popular. First, they need to invest in finding ways to automate the flow of information and standardise the storage, transmission and viewing of information, in order to improve interoperability of the systems.

Dr Kalyanpur adds that standards like HL7 (Health Level Seven) and DICOM (Digital Imaging and Communications in Medicine) hold a lot of promise.

Telepresence: putting minds together

Telepresence, long thought of as a high-end videoconferencing solution for corporates, is now turning into a telemedicine ally. Overall, it is proving to be more useful now since telepresence is now available as a ‘service’ from players such as Tata, and can be used without hefty initial investments.

Telepresence sessions prove to be very helpful in long-distance expert consultations. The consulting expert can not only view the patient’s medical records in great detail and high resolutions but also see and talk to the patient ‘in person.’ It instils more confidence in the patient, and helps the doctor diagnose and offer her opinions more confidently. Advanced telepresence systems also enable doctors to oversee or share opinions during complex surgeries.

More recently, foreign universities have started using telepresence for live demonstrations. Students can watch the surgery in great detail, just as if they are in the operation theatre. Yet, they would not be disturbing the surgeon at work.

Telepresence systems are even being used for advanced research where

geographically-separated researchers would need to see experiments or diagrams as if they were next to each other. Research networks such as Internet2 are working with telepresence providers to create futuristic systems, which would provide researchers with an immersive environment such as medical clinic configuration or lab.

However, the general opinion is that telepresence would be more useful if it could be set up in an *ad-hoc* fashion without too much advance notice to the service provider.

Health on the ‘cloud’

The highlight of a recent announcement by Verizon was the offering of telemedicine-based healthcare as a managed service, right from the equipment up to mobile integration and videoconferencing solutions. Apart from PC-based solutions, the company is also working with certain hospitals to provide specialist care using mobile-based videoconferencing. Such managed services would make it much easier and faster for hospitals to deploy telemedicine services to not just the urban population but even remote areas.

Medweb, a well-known telemedicine provider, has also started offering cloud services. The hosted solution is an alternative to costly, redundant infrastructure investments. The telemedicine cloud includes a 30-user licence covering one clinical specialty and